SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME: **ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA0101**

COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.*(BTL 3)
Assessment Tool Weightage: 15%

Total Marks: 100 **Time: 60 Mins**
No. of Questions: 5

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I Shaik Mubarak(192425194)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

ANSWER ANY ONE OF THE FOLLOWING

Q.No	Use Case Scenario	
1	**Smart Healthcare Diagnosis**	A hospital uses an AI system to identify possible diseases from patient symptoms. Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention. Tasks: (i) Identify facts, entities, predicates, variables, and constants. (ii) Represent the knowledge using Propositional Logic. (iii) Represent the knowledge using FOL with suitable quantifiers. (iv) Perform unification for two relevant predicates. (v) Apply resolution to prove whether Ravi has a viral infection. (vi) Show the complete inference sequence and final conclusion.
2	**Student Academic Advising**	A university AI system recommends academic support based on attendance and examination performance. Priya has attendance below 75% and has failed two courses. Students with low attendance require attendance counseling. Students who fail two or more courses require academic counseling. Students requiring both types of counseling need academic support. Tasks: (i) Identify the knowledge components. (ii) Convert the statements into Propositional Logic. (iii) Develop FOL representations using predicates and quantifiers. (iv) Identify suitable substitutions and perform unification. (v) Use resolution to prove whether Priya requires academic support. (vi) Explain the reasoning steps leading to the conclusion.
3	**Intelligent Loan Approval**	A bank wants to use an AI knowledge-based system for preliminary loan approval. Arun has stable employment, sufficient income, a high credit score, and good repayment history. Customers with stable employment and sufficient income qualify for preliminary approval. Customers with a high credit score and preliminary approval are classified as low-risk. Low-risk customers with good repayment history can be recommended for loan approval. Tasks: (i) Identify entities, facts, predicates, and rules. (ii) Map the scenario into Propositional Logic. (iii) Represent the rules using FOL.

		(iv) Demonstrate unification and substitution. (v) Use resolution to prove or disprove LoanApproved(Arun). (vi) Provide the final inference with justification.
4	**Vehicle Fault Diagnosis**	An automobile service center uses an expert system to diagnose vehicle faults. Car101 does not start and its headlights are dim. Vehicles that do not start and have dim headlights may have a battery problem. Vehicles with a battery problem require battery inspection. Vehicles requiring battery inspection are marked for service. Tasks: (i) Extract the facts and rules from the scenario. (ii) Represent them using Propositional Logic. (iii) Develop suitable FOL predicates and rules. (iv) Perform unification for the vehicle-related predicates. (v) Apply resolution to determine whether Car101 has a battery problem. (vi) Derive whether Car101 should be marked for service and explain the inference.
5	**Cybersecurity Threat Detection**	An organization uses an AI system to identify cyber threats. User101 has five failed login attempts and logged in from an unusual location. Multiple failed login attempts indicate a possible brute-force attack. A brute-force attack combined with an unusual login location indicates suspicious activity. Suspicious activity combined with unauthorized access to confidential files indicates a possible security breach. Tasks: (i) Identify the facts, entities, predicates, and rules. (ii) Map the knowledge into Propositional Logic. (iii) Represent the rules using FOL. (iv) Demonstrate unification and substitution. (v) Apply resolution to determine whether User101 is involved in a possible security breach. (vi) Present the complete reasoning chain and final inference.

EVALUATION RUBRICS:

Criterion	**Weightage (%)**	**Excellent**	**Good**	**Average**	**Poor**
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants & Quantifiers	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives & Formula Construction	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.

		construct valid logical formulas.			
Inference & Logical Reasoning	15%	13–15% – Performs accurate step-by-step reasoning and derives valid conclusions from the given knowledge.	10–12% – Reasoning is mostly correct with minor gaps.	7–9% – Provides partial reasoning with some incorrect conclusions.	0–6% – Reasoning is incorrect, unsupported, or missing.
Unification & Substitution	10%	9–10% – Correctly identifies substitutions and Most General Unifier (MGU) and applies them accurately.	7–8% – Correctly performs most unification steps with minor errors.	5–6% – Demonstrates basic understanding but has several errors.	0–4% – Unable to perform unification correctly.
Resolution / Proof Construction	10%	9–10% – Correctly applies resolution and presents a clear, logically valid proof of the query.	7–8% – Resolution is mostly correct with minor errors.	5–6% – Shows partial understanding but proof contains gaps.	0–4% – Resolution is incorrect or not demonstrated.
Clarity, Mapping Diagram & Presentation	5%	5% – Provides a clear, well-organized mapping/logic diagram with accurate notation and excellent presentation.	4% – Presentation is clear with minor formatting or diagram issues.	3% – Basic presentation with limited clarity.	0–2% – Poorly organized, unclear, or incomplete presentation.

1.SMART HEALTHCARE DIAGNOSIS SYSTEM USING KNOWLEDGE REPRESENTATION AND LOGICAL INFERENCE

1. Introduction

Artificial Intelligence can assist healthcare professionals by analyzing patient symptoms and identifying possible diseases. A knowledge-based healthcare diagnosis system stores medical knowledge as facts and rules and uses logical inference to derive conclusions.In this case study, an AI system analyzes the symptoms of a patient named Ravi. Ravi has fever, cough, and body pain. The knowledge base contains rules stating that patients with fever and cough may have flu, and patients with flu and body pain may have a viral infection.

The system uses Propositional Logic, First-Order Logic (FOL), predicates, variables, constants, quantifiers, unification, substitution, and resolution to determine whether Ravi has a viral infection.

Given Information

- Ravi has fever.
- Ravi has cough.
- Ravi has body pain.
- Patients with fever and cough may have flu.
- Patients with flu and body pain may have viral infection.
- Patients with viral infection require medical attention.

Objective

The objective is to determine logically whether:

ViralInfection(Ravi)

can be inferred from the given facts and rules.

2. Problem Understanding and Knowledge Identification

2.1 Entities

The main entity in the scenario is:

- **Ravi** – the patient being diagnosed.

The general entity is:

- **Patient** – a person whose symptoms are being analyzed.

2.2 Facts

The facts directly provided by the scenario are:

1. Ravi has fever.
2. Ravi has cough.
3. Ravi has body pain.

These can be represented as:

- Fever(Ravi)
- Cough(Ravi)
- BodyPain(Ravi)

2.3 Predicates

Predicates describe properties or conditions of a patient.

Predicate	Meaning
Fever(x)	x has fever
Cough(x)	x has cough
BodyPain(x)	x has body pain
Flu(x)	x has flu
ViralInfection(x)	x has a viral infection
MedicalAttention(x)	x requires medical attention

2.4 Constants

A constant represents a specific entity.

Ravi is the constant representing the patient.

2.5 Variable

The variable **x** represents any patient.

For example:
Fever(x) means that x has fever.
2.6 Relationships and Reasoning Requirements
The knowledge base contains the following relationships:

- Fever + Cough → Flu
- Flu + Body Pain → Viral Infection
- Viral Infection → Medical Attention

The required reasoning is:

Ravi's symptoms → Flu → Viral Infection → Medical Attention

3. Logical Representation and Mapping
3.1 Propositional Logic
Propositional Logic represents complete statements using symbols.
Let:

- **P** = Ravi has fever.
- **Q** = Ravi has cough.
- **R** = Ravi has body pain.
- **S** = Ravi has flu.
- **T** = Ravi has viral infection.
- **U** = Ravi requires medical attention.

Facts
P = True
Q = True
R = True
Rules
Rule 1:
Patients with fever and cough may have flu.
P ∧ Q → S
Rule 2:
Patients with flu and body pain may have viral infection.
S ∧ R → T
Rule 3:
Patients with viral infection require medical attention.
T → U
Complete Propositional Representation
P ∧ Q → S
S ∧ R → T
T → U
Since P, Q, and R are true:
P ∧ Q = True
Therefore:
S = True
Then:
S ∧ R = True
Therefore:
T = True
Finally:
T = True
Therefore:
U = True

4. First-Order Logic with Quantifiers

First-Order Logic is more expressive than Propositional Logic because it supports variables and quantifiers.

4.1 Constants

Ravi

4.2 Variables

x represents any patient.

4.3 Predicates

- Fever(x)
- Cough(x)
- BodyPain(x)
- Flu(x)
- ViralInfection(x)
- MedicalAttention(x)

4.4 FOL Facts

The facts about Ravi are:
Fever(Ravi)
Cough(Ravi)
BodyPain(Ravi)

4.5 FOL Rules

Rule 1 – Fever and Cough imply Flu

For every patient:
$\forall x\ [(Fever(x) \land Cough(x)) \rightarrow Flu(x)]$
Meaning:
For every patient x, if x has fever and cough, then x has flu.

Rule 2 – Flu and Body Pain imply Viral Infection

$\forall x\ [(Flu(x) \land BodyPain(x)) \rightarrow ViralInfection(x)]$
Meaning:
For every patient x, if x has flu and body pain, then x has a viral infection.

Rule 3 – Viral Infection requires Medical Attention

$\forall x\ [ViralInfection(x) \rightarrow MedicalAttention(x)]$
Meaning:
For every patient x, if x has a viral infection, then x requires medical attention.

4.6 Why Universal Quantification is Used

The rules are general medical rules that can apply to any patient, not only Ravi.
Therefore, the universal quantifier **∀x** is appropriate.

5. Logical Connectives and Formula Construction

The following logical connectives are used in the knowledge representation.

Symbol	Meaning	Example
∧	AND	Fever(x) ∧ Cough(x)
∨	OR	Fever(x) ∨ Cough(x)
¬	NOT	¬Fever(x)
→	Implies	Fever(x) → Flu(x)
↔	If and only if	A ↔ B

5.1 AND

Both symptoms are required:

Fever(x) ∧ Cough(x)

This means the patient must have both fever and cough.

5.2 Implication

The rule:

Fever(x) ∧ Cough(x) → Flu(x)

means that fever and cough together imply flu.

5.3 NOT

The negation of fever is:
$\neg Fever(x)$
which means that the patient does not have fever.

5.4 OR

An alternative condition can be represented as:
$Fever(x) \lor Cough(x)$
meaning the patient has fever or cough.

5.5 Complete Knowledge Representation

The main logical rules are:

$$\forall x\ [(Fever(x) \land Cough(x)) \rightarrow Flu(x)]$$
$$\forall x\ [(Flu(x) \land BodyPain(x)) \rightarrow ViralInfection(x)]$$
$$\forall x\ [ViralInfection(x) \rightarrow MedicalAttention(x)]$$

6. Knowledge Base

The knowledge base contains facts and rules.

6.1 Facts

- **F1:** Fever(Ravi)
- **F2:** Cough(Ravi)
- **F3:** BodyPain(Ravi)

6.2 Rules

- **R1:** $Fever(x) \land Cough(x) \rightarrow Flu(x)$
- **R2:** $Flu(x) \land BodyPain(x) \rightarrow ViralInfection(x)$
- **R3:** $ViralInfection(x) \rightarrow MedicalAttention(x)$

6.3 Knowledge Base Structure

```
                  KNOWLEDGE BASE
                        |
              +-----------+-----------+
                  |               |
                FACTS           RULES
                  |               |
          Fever(Ravi)         Fever(x) ∧ Cough(x)
             Cough(Ravi)              → Flu(x)
         BodyPain(Ravi)        Flu(x) ∧ BodyPain(x)
                    → ViralInfection(x)
                      ViralInfection(x)
                   → MedicalAttention(x)
```

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7. Unification and Substitution

Unification is the process of finding substitutions that make two logical expressions identical.

7.1 Unification Example 1

Consider:
Fever(x)
and
Fever(Ravi)
To make the predicates identical:
$x = Ravi$
Therefore, the substitution is:
$\theta = \{x/Ravi\}$
After substitution:
Fever(Ravi)
Hence, unification succeeds.

7.2 Unification Example 2

Consider:

Cough(x)
and
Cough(Ravi)
The required substitution is:
$\theta = \{x/Ravi\}$
After substitution:
Cough(Ravi)
Therefore, the predicates successfully unify.

7.3 Unification with a Rule

Consider Rule 1:
Fever(x) ∧ Cough(x) → Flu(x)
Given:
Fever(Ravi)
and
Cough(Ravi)
Using:
$\theta = \{x/Ravi\}$
the rule becomes:
Fever(Ravi) ∧ Cough(Ravi) → Flu(Ravi)
Since both premises are true:
Flu(Ravi) is derived.

8. Resolution and Proof Construction

Resolution is a logical inference method used to prove whether a conclusion follows from a set of facts and rules.
We want to prove:
ViralInfection(Ravi)

8.1 Convert Rules into Clause Form

Rule 1

Original:
Fever(x) ∧ Cough(x) → Flu(x)
Convert implication:
¬(Fever(x) ∧ Cough(x)) ∨ Flu(x)
Using De Morgan's law:
¬Fever(x) ∨ ¬Cough(x) ∨ Flu(x)
For Ravi:
¬Fever(Ravi) ∨ ¬Cough(Ravi) ∨ Flu(Ravi)

Rule 2

Original:
Flu(x) ∧ BodyPain(x) → ViralInfection(x)
Clause form:
¬Flu(x) ∨ ¬BodyPain(x) ∨ ViralInfection(x)
For Ravi:
¬Flu(Ravi) ∨ ¬BodyPain(Ravi) ∨ ViralInfection(Ravi)

8.2 Known Facts

Fever(Ravi)
Cough(Ravi)
BodyPain(Ravi)

8.3 Resolution Step 1

Start with:
¬Fever(Ravi) ∨ ¬Cough(Ravi) ∨ Flu(Ravi)
Resolve with:
Fever(Ravi)
Result:

¬Cough(Ravi) ∨ Flu(Ravi)
Resolve with:
Cough(Ravi)
Result:
Flu(Ravi)
Therefore:
Flu(Ravi) is proven.

8.4 Resolution Step 2

Use Rule 2:
¬Flu(Ravi) ∨ ¬BodyPain(Ravi) ∨ ViralInfection(Ravi)
We already proved:
Flu(Ravi)
Resolving gives:
¬BodyPain(Ravi) ∨ ViralInfection(Ravi)
We know:
BodyPain(Ravi)
Resolving again gives:
ViralInfection(Ravi)
Therefore:
ViralInfection(Ravi) is proven.

9. Resolution by Refutation

Resolution can also be demonstrated using refutation.
Query
ViralInfection(Ravi)
Assume the opposite:
¬ViralInfection(Ravi)
We have already derived:
ViralInfection(Ravi)
Therefore, combining:
ViralInfection(Ravi)
with:
¬ViralInfection(Ravi)
produces a contradiction:
□
The empty clause □ represents a contradiction.
Therefore, the assumption is false and:
ViralInfection(Ravi)
is logically proven.

10. Inference and Logical Reasoning – 15%

The complete inference sequence is:

Step 1 – Initial Facts

The system receives:

Fever(Ravi)
Cough(Ravi)
BodyPain(Ravi)

Step 2 – Apply Rule 1

Rule:
Fever(x) ∧ Cough(x) → Flu(x)
Substitution:
x = Ravi
Therefore:
Fever(Ravi) ∧ Cough(Ravi) → Flu(Ravi)

Since both facts are true:
Flu(Ravi)
is derived.

Step 3 – Apply Rule 2

Rule:
Flu(x) ∧ BodyPain(x) → ViralInfection(x)
Substitution:
x = Ravi
Therefore:
Flu(Ravi) ∧ BodyPain(Ravi) → ViralInfection(Ravi)
Both conditions are true.
Therefore:
ViralInfection(Ravi)
is derived.

Step 4 – Apply Rule 3

Rule:
ViralInfection(x) → MedicalAttention(x)
Substitution:
x = Ravi
Therefore:
MedicalAttention(Ravi)
is derived.

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11. Complete Inference Sequence

```
          INITIAL FACTS
                |
       +-------+-------+
       |       |       |
    Fever   Cough   Body Pain
            |       |
            +---+---+
                |
                v
         RULE 1 APPLIED
                |
                v
            Flu(Ravi)
                |
           +---------+
                |
          BodyPain(Ravi)
                |
                v
         RULE 2 APPLIED
                |
                v
       ViralInfection(Ravi)
                |
                v
         RULE 3 APPLIED
                |
                v
      MedicalAttention(Ravi)
```

Inference Chain in One Line

Fever(Ravi) + Cough(Ravi)
↓
Flu(Ravi)
↓
Flu(Ravi) + BodyPain(Ravi)
↓
ViralInfection(Ravi)
↓
MedicalAttention(Ravi)

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12. Mapping / System Architecture Diagram

The architecture of the smart healthcare diagnosis system can be represented as:

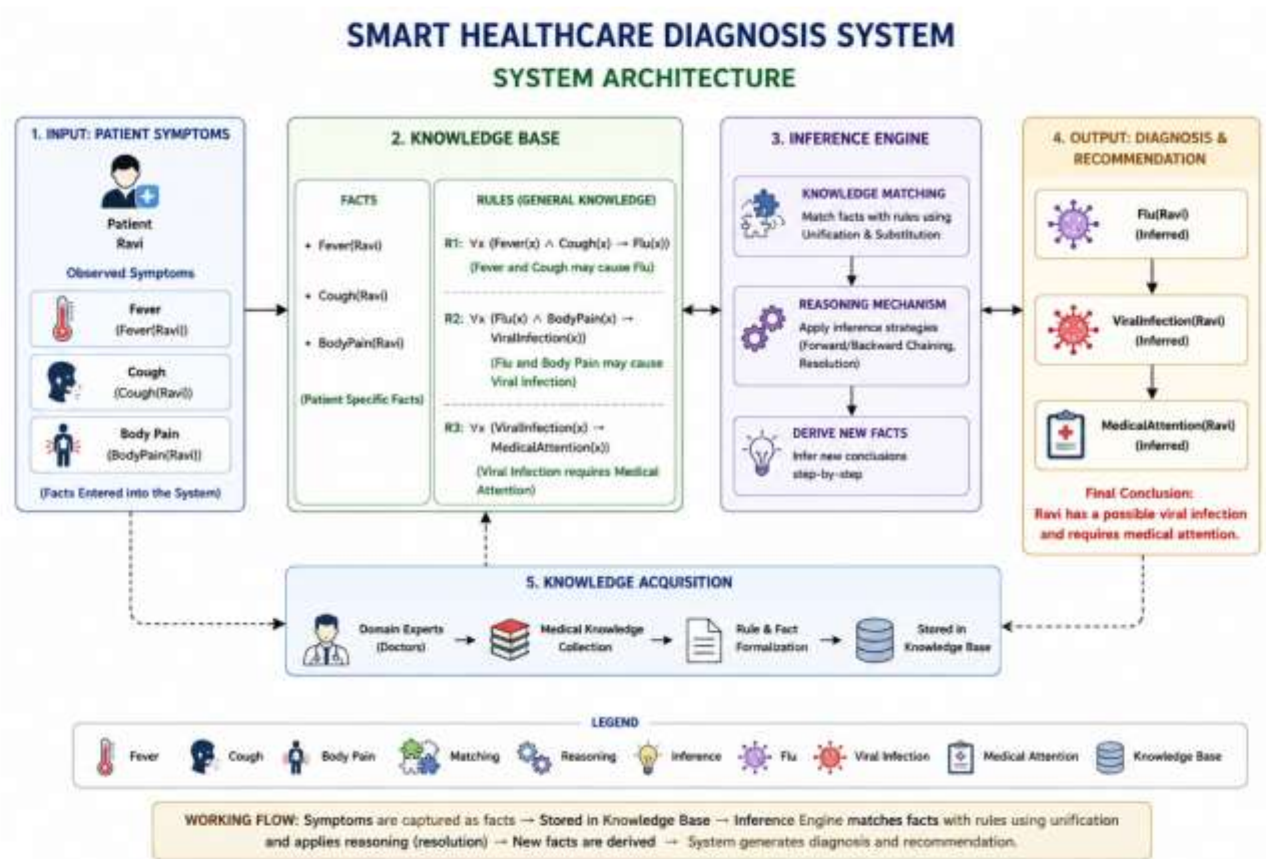

Diagram Explanation

The system first collects symptoms from the patient. These symptoms are stored as facts in the knowledge base. The knowledge base also contains general medical rules.The inference engine applies unification and logical reasoning to match the patient's facts with the rules. It derives new information such as flu and viral infection.

Finally, the system concludes that Ravi requires medical attention.

13. Final Conclusion

The Smart Healthcare Diagnosis System demonstrates how Artificial Intelligence can use knowledge representation and logical reasoning to infer a patient's possible condition.

The system starts with three facts:

- **Fever(Ravi)**
- **Cough(Ravi)**
- **BodyPain(Ravi)**

Using the first rule:

Fever(Ravi) ∧ Cough(Ravi) → Flu(Ravi)

the system derives:

Flu(Ravi)
Using the second rule:
Flu(Ravi) ∧ BodyPain(Ravi) → ViralInfection(Ravi)
the system derives:
ViralInfection(Ravi)
Finally, using:
ViralInfection(Ravi) → MedicalAttention(Ravi)
the system derives:
MedicalAttention(Ravi)
Final Inference

Ravi has a possible viral infection and requires medical attention.

The resolution process confirms that ViralInfection(Ravi) logically follows from the given knowledge base.
Thus, the complete reasoning chain is:
Fever + Cough → Flu
Flu + Body Pain → Viral Infection
Viral Infection → Medical Attention

This demonstrates how Propositional Logic, First-Order Logic, unification, substitution, and resolution can be integrated into an AI-based healthcare diagnosis system.
